Essentials of Artificial Intelligence

Group Assignment

Intake 41 – IT3182 (IT/IS)

Intake 42 – CS 22032 (DBA) / CS22023 (CS/SE/COE)

AI-Based Intelligent Solution for a Real-World Problem

Assignment Overview

Module	Essentials of Artificial Intelligence
Duration	10 Weeks
Mode	Group Assignment
Group Type	Mixed groups with students from CS, SE, COE, DBA, IT, and IS
Assessment Stages	Stage 1: Proposal | Stage 2: Progress Review | Stage 3: Final Submission
Continuous Assessment Weight	**50% total**: Proposal 10%, Progress Review 10%, Final Submission 30%

Assignment Question

You are required to work in groups to identify a real-world problem and design an AI-based intelligent solution for that problem. The selected problem should be practical, understandable, and suitable to be addressed using basic Artificial Intelligence concepts learned in the module.

You are expected to **analyse the problem**, **explain how AI can be used to solve or support decision-making in that problem**, **design the proposed intelligent system**, and **develop a basic prototype/model**.

The final output does not need to be a fully commercial-level software product; however, it must clearly demonstrate how AI concepts are applied to solve, support, or improve decision-making in the selected problem.

Each member must have a clear responsibility, and all members must contribute to the proposal, progress review, final report, and presentation.

What You Have to Do

1. Select a Suitable Real-World Problem

Each group must select a real-world problem that can be solved or supported using AI. The topic must be feasible within 10 weeks, useful to a specific group of users, and approved by the lecturer.

2. Study the Problem Background

Students must study the selected problem by identifying what the problem is, who is affected, why it is important, how it is currently handled, limitations of the current approach, and how AI can improve the situation.

3. Define the Objectives of the AI Solution

Each group must define clear and achievable objectives. The objectives should explain what the system is expected to do, such as classify, predict, recommend, optimize, search, or support decision-making.

4. Select Suitable AI Techniques

Students must select appropriate AI techniques and justify why they are suitable for the selected problem. Students should not only name the AI technique; they must explain how the selected technique contributes to solving the problem.

5. Design the Proposed AI System

Students must prepare a system design that clearly explains the input, processing, and output of the proposed system. The design may include a system architecture diagram, flowchart, AI model diagram, knowledge representation, decision-making process, dataset structure, or user interface design.

6. Develop a Prototype, Simulation, Model, or Detailed Design

Each group must produce a final output that demonstrates the AI solution. This may be a simple web or desktop prototype, Python model, neural network model, fuzzy logic simulation, genetic algorithm simulation, search algorithm demonstration, rule-based decision-support system, or detailed AI system design with sample outputs.

7. Test the Proposed Solution

Students must test their solution using sample inputs. They should show sample input values, generated outputs, screenshots or results, whether the outputs are reasonable, and limitations of the solution.

Each group must briefly discuss ethical and social issues such as data privacy, bias, accuracy, reliability, misuse, human responsibility, and limitations of automated decisions.

Suggested Topic Areas

Area	Sample Topics
Education and Student Support	Student performance prediction; course recommendation system; examination preparation assistant; academic advisory system
Healthcare and Well-being	Health risk prediction; stress level identification using fuzzy logic; diet recommendation assistant; patient prioritization system
Smart Campus and University Management	Smart classroom allocation; campus route finder; library recommendation system; smart parking allocation
Business and Information Systems	Customer churn prediction; loan eligibility checking; fraud risk identification; sales prediction
Engineering and Smart Systems	Smart traffic signal control; delivery route optimization; energy usage recommendation; smart agriculture decision support
Security and Cyber Systems	Phishing email detection; password strength advisory; intrusion alert classification; cybersecurity risk assessment
Environment and Sustainability	Waste classification; water usage optimization; smart irrigation; disaster response route planning

Assessment Stages and Requirements

Stage 1: Proposal Submission

Due: Week 4 | **Weight:** 10%

Students must submit:

- Project title
- Names and degree programmes of group members
- Background of the selected problem
- Problem statement
- Objectives of the proposed AI solution
- Target users
- Proposed AI techniques
- Justification for selecting the AI techniques
- Expected system inputs and outputs
- Proposed tools and technologies
- Initial system diagram or workflow
- Work division among members
- Timeline for completing the work
- References

Stage 2: Progress Review

Due: Week 8 | **Weight:** 10%

Students must submit:

- Updated project title and problem statement
- Changes made after proposal feedback
- Finalized AI techniques
- System architecture or workflow diagram
- Dataset details, if applicable
- Rules, fuzzy variables, search strategy, GA design, or neural network structure
- Screenshots or sample outputs of partial implementation
- Problems faced during development
- Remaining work before final submission
- Updated contribution of each member

Stage 3: Final Submission and Presentation

Due: Week 10 | **Weight:** 30%

Students must submit:

- Final report
- Presentation slides and a video
- Prototype / source code / simulation / model / system design
- Dataset or sample data, if applicable
- Individual contribution sheet
- References

Final Report Structure

- Title page
- Abstract
- Introduction
- Problem background
- Problem statement
- Objectives
- Selected AI techniques
- Justification of AI techniques
- System design
- Implementation details
- Testing and results
- Limitations & Discussion
- Ethical and social considerations
- Conclusion
- References

- Individual contribution table

Sample Topic-to-Technique Mapping

Topic	Possible AI Technique
Campus route finder	Search algorithms
Traffic light control	Fuzzy logic
Timetable optimization	Genetic algorithm
Student performance prediction	Neural network / classification model
Phishing email detection	Rule-based reasoning / neural network
Smart irrigation system	Fuzzy logic
Waste image classification	Neural network image classification
Customer churn prediction	Neural network classification model

Important Instructions for Students

- The topic must be approved by the lecturer.
- The project must be completed within 10 weeks.
- Each group should apply at least three AI-related concepts.
- Students must clearly explain the AI technique, not only implement it.
- All group members must participate in the final presentation.
- Individual contribution must be clearly stated.
- Proper references must be included.
- Plagiarism and unacknowledged use of AI-generated content are not allowed.
- The final system should show AI-based reasoning, prediction, classification, optimization, or decision-making.